Dr. Jared Jeyaretnam

Employment

2025–now **Research Fellow**, *University of Nottingham*
I am studying non-equilibrium quantum many-body physics from a circuit perspective, including open systems, jointly supervised by Adam Gammon-Smith and Juan Garrahan. I make use of both analytical and numerical techniques, including tensor networks, exact diagonalisation, and perturbation theory.

2023–2025 **Research Fellow**, *University of Leeds*
Working in the Theory Group with Zlatko Papić, I researched out-of-equilibrium phenomena such as Hilbert space fragmentation in lattice gauge theories and many-body localisation using diverse numerical techniques such as exact diagonalisation, tensor networks, perturbation theory, Clifford circuits, and the real-space renormalisation group.

Publications

In prep. **Non-equilibrium steady states of corner-driven two-dimensional tight-binding models**, first author
Theory companion paper to "*Quantum trajectory simulation of two-dimensional non-equilibrium steady states...*" (see below), studying hydrodynamics, metastability, and long-lived oscillations in this same open quantum system. In particular, we observe and prove the existence of a decoherence-free subspace due to certain eigenstates of the tight-binding Hamiltonian, and show that it is stable to perturbations.

In prep. **Role of discrete levels in disorder-free localization: a quantum simulation perspective**, first author
We investigate a translationally-invariant model which maps to an ensemble of mixed-field Ising models with discrete bond disorder, which was studied in a recent Google Quantum AI experiment simulating lattice gauge theories. We show that evidence of localisation for two-level disorder is weak, but signatures of a phase akin to conventional many-body localisation becomes much more robust for four or more levels, with additional approximate fragmentation.

2026 **Quantum trajectory simulation of two-dimensional non-equilibrium steady states with a trapped ion quantum processor**, arXiv:2605.08350 [quant-ph]
We experimentally realise quantum trajectories for a two-dimensional system of (interacting) particles – hard-core bosons or fermions – undergoing stochastic driving at a source and drain on opposite corners of a lattice. The particle statistics, presence of interactions, and introduction of a magnetic field produce measurable effects on the steady state, highlighting the rich physics in this corner driven two-dimensional setup.

2026 **Ergodicity breaking in matrix-product-state effective Hamiltonians**, arXiv:2603.26870 [cond-mat.str-el], first author
Here we demonstrate that the density-matrix renormalization group (DMRG) effective Hamiltonian, an object routinely used to variationally approximate ground states, encodes detailed information about the dynamics far from equilibrium. The spectrum of the effective Hamiltonian is shown to capture the transition from thermal to many-body localized regimes, including spatially resolved probes of ergodic bubbles. Furthermore, the same approach also captures weak ergodicity breaking associated with quantum many-body scars.

2025 **Hilbert space fragmentation at the origin of disorder-free localization in the lattice Schwinger model**, Commun. Phys. **8**, 172 (2025), first author, featured selection
Recent works have reported potential disorder-free localization in the lattice Schwinger model. Using degenerate perturbation theory and numerical simulations based on exact diagonalization and matrix product states, we identify the origin of a claimed ultraslow growth of entanglement as due to an approximate Hilbert space fragmentation and an emergent dynamical constraint on particle hopping.

2023 **Ergodicity breaking and stabilisation of quantum order**, PhD Thesis

2023 **Renormalisation view on resonance proliferation between many-body localised phases**, Phys. Rev. B **108**, 094205, first author
In this work, we study the statistical properties of many-body resonances in a disordered interacting Ising chain – which can host symmetry protected topological order – using a Clifford circuit encoding of the real space renormalisation group. We show that both MBL phases present remain stable to resonances, but in the vicinity of the transition between them localisation is destabilised by resonance proliferation.

2021 **Quantum scars and bulk coherence in a symmetry-protected topological phase**, Phys. Rev. B **104**, 014424, first author
Quantum many-body scars provide a novel mechanism for enhancing coherence of weakly entangled states; while coherent edge modes in certain symmetry protected topological (SPT) phases can persist away from the ground state. We uncover many-body scars and their implications on bulk coherence in such an SPT phase, shedding light on their role in preserving SPT order at finite temperature and the possibility of coherent bulk dynamics in models with SPT order beyond long-lived edge modes.

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Education

2019–2023 **PhD, Condensed Matter Physics**, *University College London*, with Arijeet Pal
Studied out-of-equilibrium phenomena in quantum mechanics, in particular quantum scars and many-body localisation, focusing on systems hosting symmetry-protected topological (SPT) order and their stabilisation at finite temperature via the mechanism of ergodicity breaking.

2015–2019 **BA, MSci Natural Sciences**, *University Of Cambridge*, Senior Scholar, First Class
Top-ranked First in third year ("Part II"), Part II Hartree and Clerk Maxwell Prizewinner. Scored 95% overall in second-year Maths. Very high Firsts in all four years. Master's project studied edge states and topological order in an eightfold quasicrystal; also used Monte-Carlo simulations to study spin ice.

Talks and Posters

Apr. 2026 **Poster**, *Ergodicity breaking in matrix-product-state effective Hamiltonians*, SimQuDyn (Munich)

Apr. 2025 **Talk**, *Hilbert Space Fragmentation and Disorder-Free Localisation in a Lattice Gauge Theory*, Nottingham Symposium on Quantum Systems
Also presented as a poster at SimQuDyn *in Munich (Apr. 2025), and at* New frontiers in out-of-equilibrium quantum many-body dynamics *at MPI-PKS (Aug. 2025)*

Sep. 2024 **Posters**, *Quantum Circuit Analysis of an MBL phase transition* and *Does Many-Body Localization exist in U(1) lattice gauge theories?*, Localisation: Emergent Trends and Novel Platforms (MPI-PKS)

Jun. 2024 **Talk**, *Localisation and Hilbert Space Fracture in Lattice Gauge Theories*, Bridging the Physics and Mathematics of Quantum Many Body Chaos (Helsinki)

Jan. 2024 **Poster**, *Quantum Circuit Analysis of an MBL phase transition*, International Quantum Tensor Network (Glasgow)

Feb. 2023 **Poster**, *Renormalisation view on resonance proliferation*, Quantum Information Processing (Ghent)

Software Contributions

2025 **Makie.jl**, *Github*, pull request
Fixed a bug in axis scaling logic for plots.

2025 **Julia for VSCode**, pull request
Fixed a bug with profiling in Julia 1.12.

2024 **Numba**, pull request
Added support for compiling additional `numpy` functions related to set operations, which are useful for certain numerical methods in quantum mechanics (such as manipulating Pauli strings).

Non-Research Experience

Teaching

2023-2025 **MSci and BSc projects**, *University of Leeds*
Assisted with supervision of Master's and Bachelor's projects, helping familiarise students with the material and develop their codebase for research.

2023-2025 **Physics tutor**, *University of Leeds*
Volunteered in an after-school homework club for A-level students from local schools.

2020-2022 **Postgraduate Teaching Assistant**, *University College London*
Led and planned a series of problem-solving tutorials on mathematical methods. Assisted with classes on Python and scientific computation.

Science Communication

2023-2026 **Seminar series**, *University of Leeds/Nottingham*
Organise seminars in collaboration with Loughborough, Leeds, and Nottingham universities, including inviting speakers, hosting and recording seminars, and maintaining a website.

2016–2023 **Committee Member (various roles)**, *CHaOS*
Cambridge Hands on Science, or CHaOS, is a student science outreach society which aims to inspire children to pursue STEM with fun, hands on experiments and demonstrations. Organised venues and accomodation for our summer roadshow. Managed teams of student demonstrators at numerous CHaOS events, many of whom were inexperienced. Ensured all of our ∼120 experiments were safety checked in time for deadlines as Safety Officer.

Non-Academic Interests

Science Outreach: See CHaOS, above

Climbing: Enthusiastic rock climber, both indoors and outdoors!

Travel: Visited over 45 countries in four continents – confident travelling anywhere and interacting with people of any culture